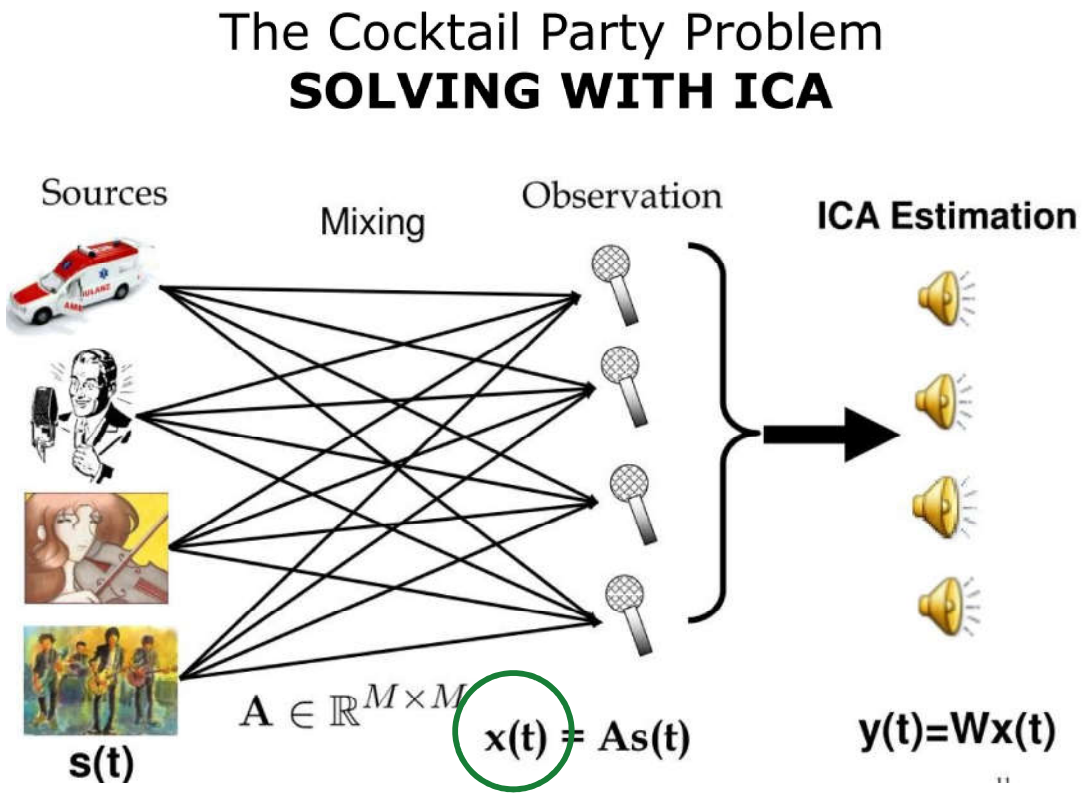
Independent Component Analysis (ICA)

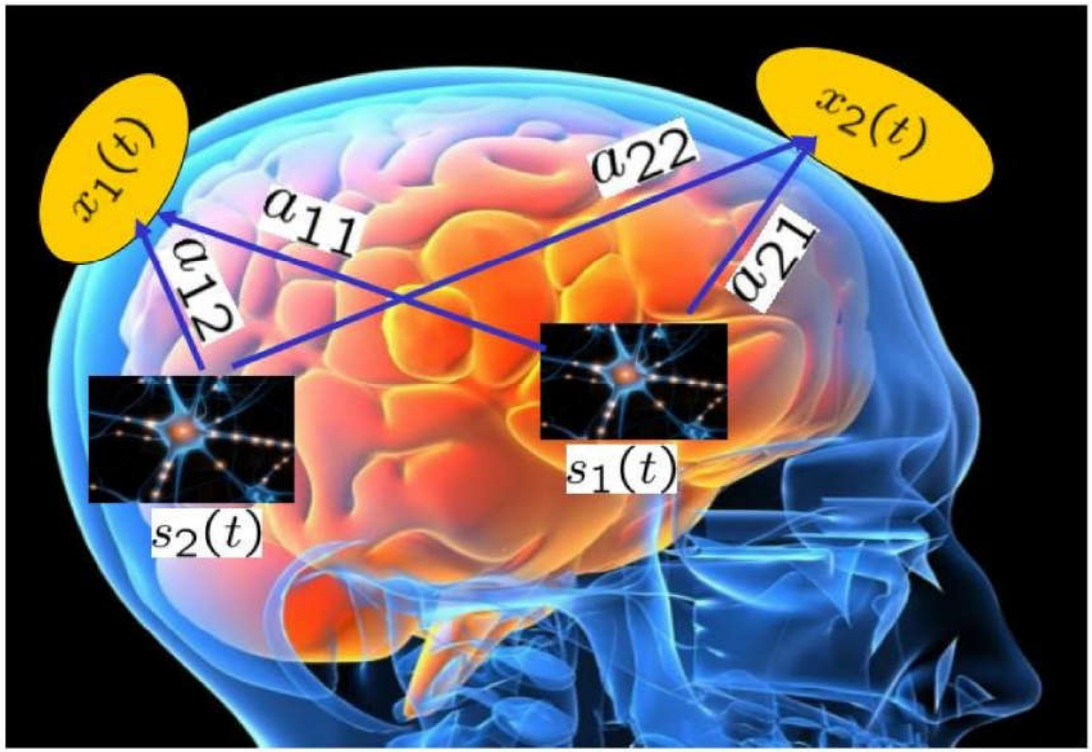
Here, *d* speakers are speaking simultaneously at a party, and any microphone placed in the room records only an overlapping combination of the *d* speakers' voices.
We have *d* different microphones placed in the room, and because each microphone is a different distance from each of the speakers, it records a different combination of the speakers' voices.
Using these microphone recordings, can we **separate out the original *d* speakers' speech signals**?

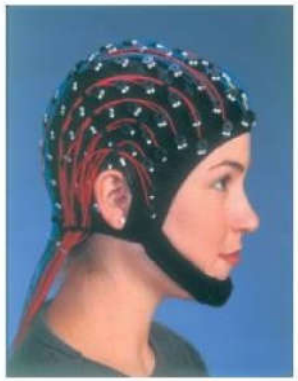


Independent Component Analysis

$$x_1(t) = a_{11}s_1(t) + a_{12}s_2(t)$$
$$x_2(t) = a_{21}s_1(t) + a_{22}s_2(t)$$

Goal: Estimate $\{s_i(t)\}$, (and also $\{a_{ij}\}$)

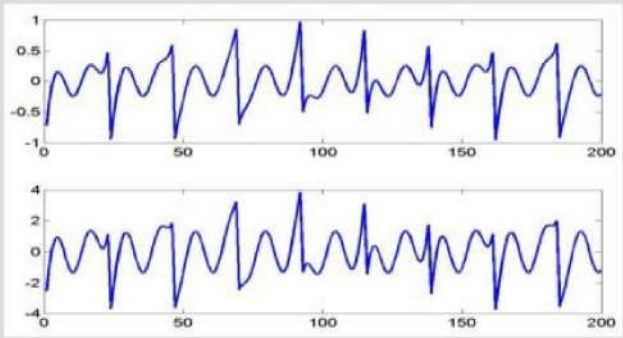




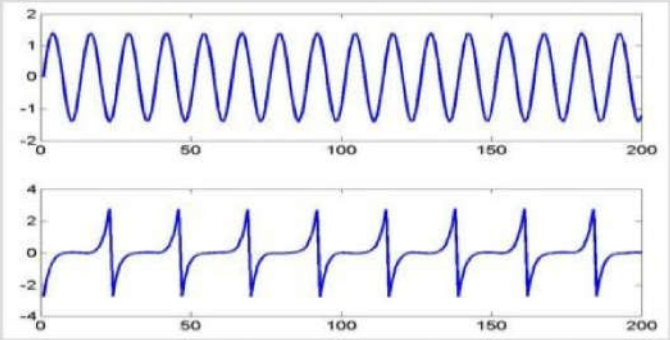
Independent Component Analysis

$$x_1(t) = a_{11}s_1(t) + a_{12}s_2(t)$$
$$x_2(t) = a_{21}s_1(t) + a_{22}s_2(t)$$

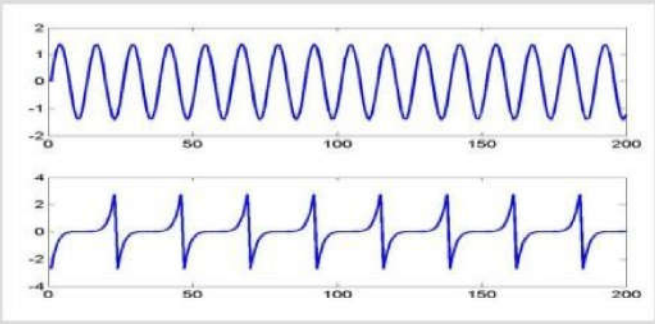
Model



Observations (Mixtures)



ICA estimated signals



original signals

Independent Component Analysis

Model

$$x_1(t) = a_{11}s_1(t) + a_{12}s_2(t)$$
$$x_2(t) = a_{21}s_1(t) + a_{22}s_2(t)$$

We observe

$$\begin{pmatrix} x_1(1) \\ x_2(1) \end{pmatrix}, \begin{pmatrix} x_1(2) \\ x_2(2) \end{pmatrix}, \ldots, \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix}$$

We want

$$\begin{pmatrix} s_1(1) \\ s_2(1) \end{pmatrix}, \begin{pmatrix} s_1(2) \\ s_2(2) \end{pmatrix}, \ldots, \begin{pmatrix} s_1(t) \\ s_2(t) \end{pmatrix}$$

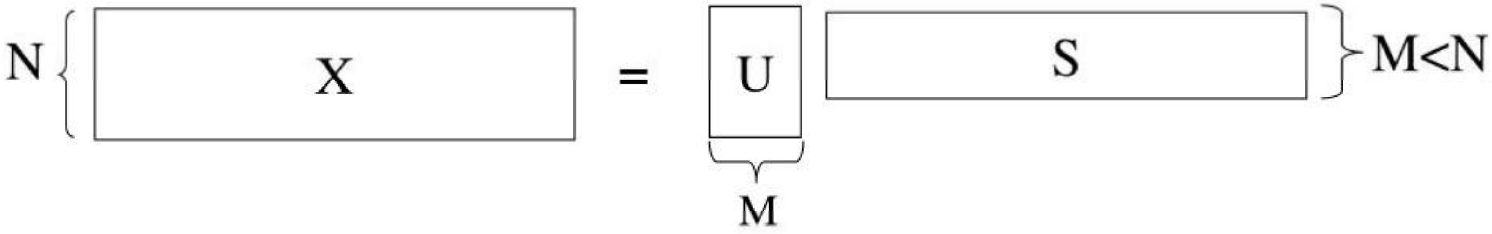
But we don't know $\{a_{ij}\}$, nor $\{s_i(t)\}$

Goal: Estimate $\{s_i(t)\}$, (and also $\{a_{ij}\}$)

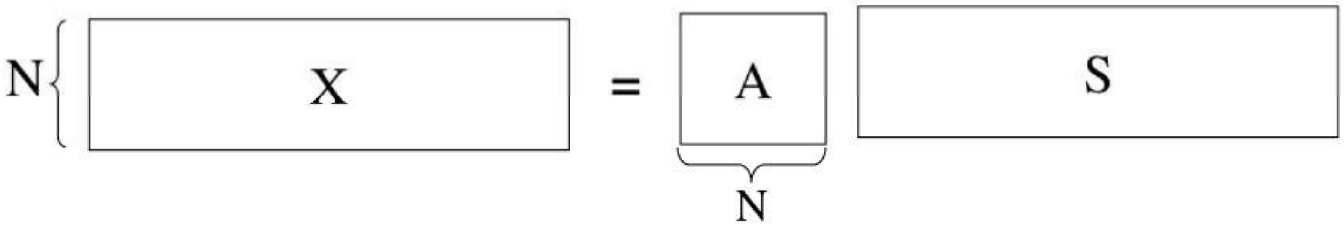
ICA vs PCA, **Similarities**

- Perform linear transformations
- Matrix factorization

PCA: ***low rank*** matrix factorization for ***compression***



ICA: ***full rank*** matrix factorization to ***remove dependency*** between the rows



ICA vs PCA, **Differences**

- PCA **does** compression
  - M<N
- ICA does **not** do compression
  - same # of features (M=N)

- PCA just removes correlations, **not** higher order dependence
- ICA removes correlations, **and** higher order dependence

- PCA: $\mathbf{X=US, U^TU=I}$
- ICA: $\mathbf{X=AS}$

- PCA: some components are **more important** than others (based on eigenvalues)
- ICA: components are **equally important**

Some ICA Applications

STATIC

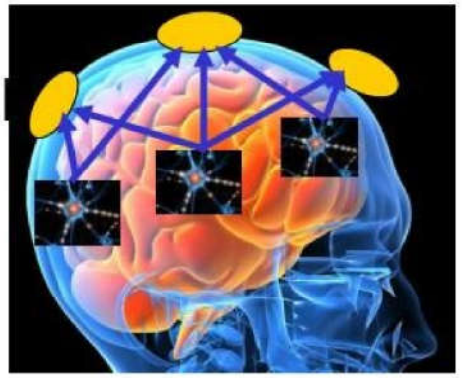
- Image denoising
- Microarray data processing
- Decomposing the spectra of galaxies
- Face recognition
- Facial expression recognition
- Feature extraction
- Clustering
- Classification

TEMPORAL

- Medical signal processing – fMRI, ECG, EEG
- Brain Computer Interfaces
- Modeling of the hippocampus, place cells
- Modeling of the visual cortex
- Time series analysis
- Financial applications
- Blind deconvolution

ICA Application, Removing Artifacts from EEG

- EEG ~ *Neural cocktail party*
- Severe ***contamination*** of EEG activity
  - eye movements
  - blinks
  - muscle
  - heart, ECG artifact
  - vessel pulse
  - electrode noise
  - line noise, alternating current (60 Hz)



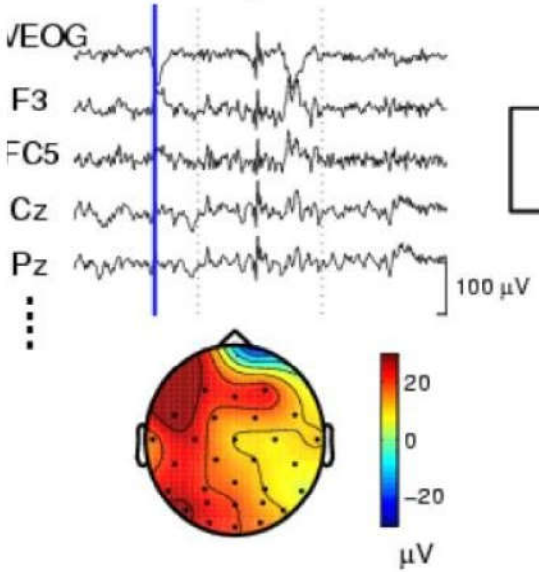
- ICA can improve signal
  - effectively ***detect, separate and remove*** activity in EEG records from a wide variety of artifactual sources. (Jung, Makeig, Bell, and Sejnowski)

- ICA weights help find **location** of sources



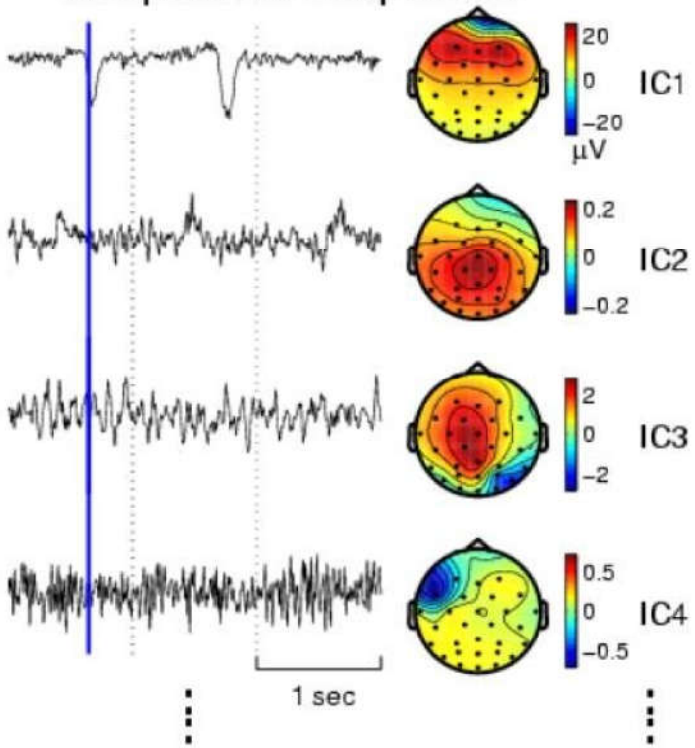
ICA decomposition

EEG Scalp Channels



unmixing
(W)

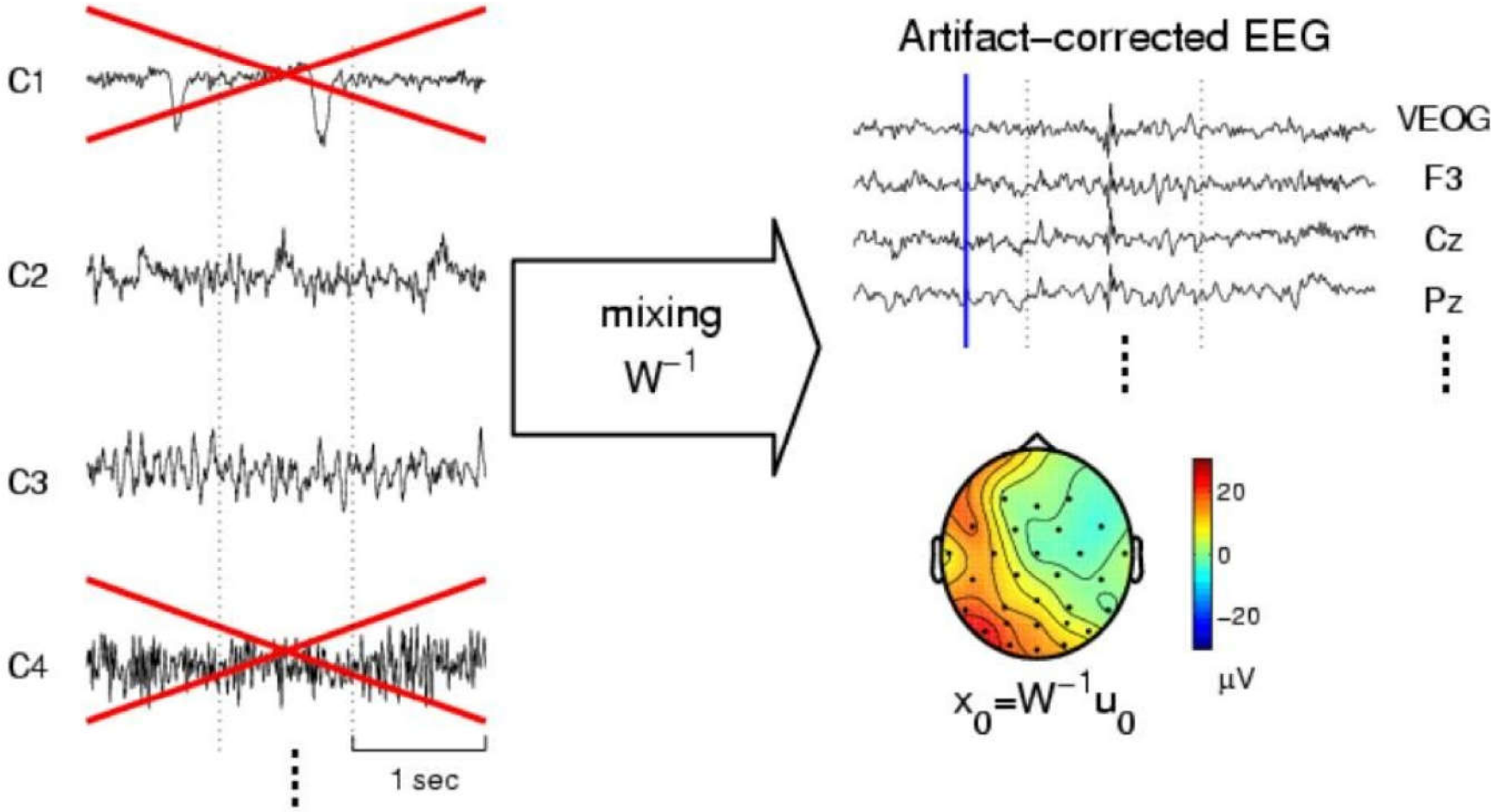
Independent Components

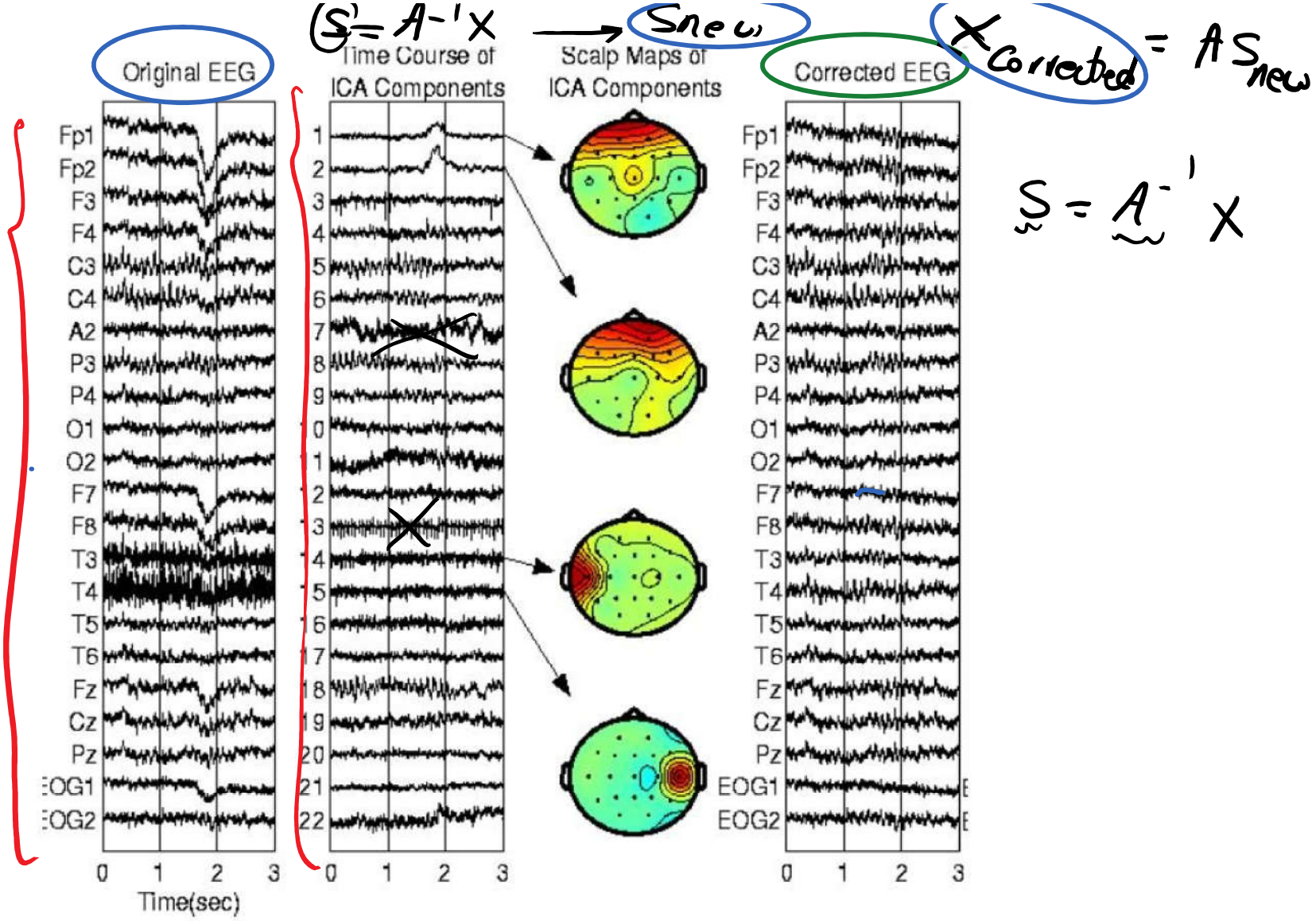


activations
(u=WX)

scalp maps
(W^{-1})

Summed Projection of Selected Components




S = A⁻¹X
Snew
X Corrected = A Snew
Original EEG
Time Course of ICA Components
Scalp Maps of ICA Components
Corrected EEG
Fp1
Fp2
F3
F4
C3
C4
A2
P3
P4
O1
O2
F7
F8
T3
T4
T5
T6
Fz
Cz
Pz
EOG1
EOG2
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
0
1
2
3
Time(sec)
S = A⁻¹ X

ICA for Image Denoising
(Hoyer, Hyvarinen)



original



noisy



Wiener filtered



median filtered

ICA denoised



To formalize this problem, we imagine that there is some data $s \in \mathbb{R}^d$ that is generated via d independent sources. What we observe is

$$x = As,$$

where A is an unknown square matrix called the **mixing matrix**. Repeated observations gives us a dataset $\{x^{(i)}; i = 1, \ldots, n\}$, and our goal is to recover the sources $s^{(i)}$ that had generated our data ($x^{(i)} = As^{(i)}$).

In our cocktail party problem, $s^{(i)}$ is an d-dimensional vector, and $s_j^{(i)}$ is the sound that speaker j was uttering at time i. Also, $x^{(i)}$ in an d-dimensional vector, and $x_j^{(i)}$ is the acoustic reading recorded by microphone j at time i.

$W = A^{-1}$ $\longrightarrow$ $s^{(i)} = Wx^{(i)}$

Unmixing matrix

$$W = \begin{bmatrix} w_1^T \\ \vdots \\ w_d^T \end{bmatrix} \longrightarrow s_j^{(i)} = w_j^T x^{(i)}$$

مشاهدات

بازسازی ضرایب

$w_j \in \mathbb{R}^d$

ICA Ambiguities



Permutation

Specifically, let P be any d-by-d permutation matrix. This means that each row and each column of P has exactly one "1." Here are some examples of permutation matrices:

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}; \quad P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}; \quad P = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

If z is a vector, then Pz is another vector that contains a permuted version of z's coordinates. Given only the $x^{(i)}$'s, there will be no way to distinguish between W and PW. Specifically, the permutation of the original sources is ambiguous, which should be no surprise. Fortunately, this does not matter for most applications.

ICA Ambiguities

Scaling

Further, there is no way to recover the correct scaling of the w_i's. For instance, if A were replaced with $2A$, and every $s^{(i)}$ were replaced with $(0.5)s^{(i)}$, then our observed $x^{(i)} = 2A \cdot (0.5)s^{(i)}$ would still be the same. More broadly, if a single column of A were scaled by a factor of α, and the corresponding source were scaled by a factor of $1/\alpha$, then there is again no way to determine that this had happened given only the $x^{(i)}$'s. Thus, we cannot recover the "correct" scaling of the sources. However, for the applications that we are concerned with—including the cocktail party problem—this ambiguity also does not matter. Specifically, scaling a speaker's speech signal $s_j^{(i)}$ by some positive factor α affects only the volume of that speaker's speech. Also, sign changes do not matter, and $s_j^{(i)}$ and $-s_j^{(i)}$ sound identical when played on a speaker. Thus, if the w_i found by an algorithm is scaled by any non-zero real number, the corresponding recovered source $s_i = w_i^T x$ will be scaled

Sources should not be Gaussian

Are these the only sources of ambiguity in ICA? It turns out that they are, so long as the sources s_i are *non-Gaussian*. To see what the difficulty is with Gaussian data, consider an example in which $n = 2$, and $s \sim \mathcal{N}(0, I)$. Here, I is the 2x2 identity matrix. Note that the contours of the density of the standard normal distribution $\mathcal{N}(0, I)$ are circles centered on the origin, and the density is rotationally symmetric.

Now, suppose we observe some $x = As$, where A is our mixing matrix. Then, the distribution of x will be Gaussian, $x \sim \mathcal{N}(0, AA^T)$, since

$$\mathrm{E}_{s\sim\mathcal{N}(0,I)}[x] = \mathrm{E}[As] = A\mathrm{E}[s] = 0$$

$$\mathrm{Cov}[x] = \mathrm{E}_{s\sim\mathcal{N}(0,I)}[xx^T] = \mathrm{E}[Ass^TA^T] = A\mathrm{E}[ss^T]A^T = A \cdot \mathrm{Cov}[s] \cdot A^T = AA^T$$

Now, let R be an arbitrary orthogonal (less formally, a rotation/reflection) matrix, so that $RR^T = R^TR = I$, and let $A' = AR$. Then if the data had been mixed according to A' instead of A, we would have instead observed $x' = A's$. The distribution of x' is also Gaussian, $x' \sim \mathcal{N}(0, AA^T)$, since $\mathrm{E}_{s\sim\mathcal{N}(0,I)}[x'(x')^T] = \mathrm{E}[A'ss^T(A')^T] = \mathrm{E}[ARss^T(AR)^T] = ARR^TA^T = AA^T$. Hence, whether the mixing matrix is A or A', we would observe data from a $\mathcal{N}(0, AA^T)$ distribution. Thus, there is no way to tell if the sources were mixed using A and A'. There is an arbitrary rotational component in the mixing matrix that cannot be determined from the data, and we cannot recover the original sources.

اثر تبدیل خطی روی توزیع احتمال

$S \sim p_S(s)$ $\quad X = AS$ $\quad p_X(x) = ?$

$W = A^{-1} \rightarrow S = WX$ $\quad p_X(x) = p_S(Wx)\,|W|$

دترمینان

$S \in \mathbb{R}$ $\quad A \in \mathbb{R}$ $\quad X \in \mathbb{R}$ $\quad A = 2$ $\quad W = 0.5$

مثلاً $S \sim \text{Uniform}(0,1)$ $\quad X = 2S \rightarrow X \sim \text{Uniform}(0,2)$

$p_S(0.5) = 1 \rightarrow p_X(2 \times 0.5) = 1 \times |W| = 0.5$

ICA Algorithm

We are now ready to derive an ICA algorithm. We describe an algorithm by Bell and Sejnowski, and we give an interpretation of their algorithm as a method for maximum likelihood estimation. (This is different from their original interpretation involving a complicated idea called the infomax principal which is no longer necessary given the modern understanding of ICA.)

We suppose that the distribution of each source s_j is given by a density p_s, and that the joint distribution of the sources s is given by

$$p(s) = \prod_{j=1}^{d} p_s(s_j).$$

Note that by modeling the joint distribution as a product of marginals, we capture the assumption that the sources are independent. Using our formulas from the previous section, this implies the following density on $x = As = W^{-1}s$:

$$p(x) = \prod_{j=1}^{d} p_s(w_j^T x) \cdot |W|.$$

ICA Algorithm

* اگر اطلاعات قبلی راجع به منابع S داریم $p_s(s)$ را از روی آن حدس می‌زنیم.

All that remains is to specify a density for the individual sources p_s.

Recall that, given a real-valued random variable z, its cumulative distribution function (cdf) F is defined by $F(z_0) = P(z \leq z_0) = \int_{-\infty}^{z_0} p_z(z)dz$ and the density is the derivative of the cdf: $p_z(z) = F'(z)$.

$\frac{1}{1+e^{-z}}$

Thus, to specify a density for the s_i's, all we need to do is to specify some cdf for it. A cdf has to be a monotonic function that increases from zero to one. Following our previous discussion, we cannot choose the Gaussian cdf, as ICA doesn't work on Gaussian data. What we'll choose instead as a reasonable "default" cdf that slowly increases from 0 to 1, is the sigmoid function $g(s) = 1/(1+e^{-s})$. Hence $p_s(s) = g'(s)$. $= g(s)(1-g(s))$

توزیعی با میانگین صفر پیشنهاد می‌شود

$E(s) = 0$ → فرض این است که میانگین S یا صراحتاً یا خودمان صفر کردیم ...

ICA Algorithm

$g' = g(1-g) \longrightarrow g'' = g'(1-g) - gg' = g' - 2gg'$

The square matrix W is the parameter in our model. Given a training set $\{x^{(i)}; i = 1, \ldots, n\}$, the log likelihood is given by $\frac{d}{dw}(\ln g'(w^T x)) = \underbrace{g''}_{g'(1-2g)} x^T$

$$\ell(W) = \sum_{i=1}^{n} \left(\sum_{j=1}^{d} \log g'(w_j^T x^{(i)}) + \log |W| \right).$$

$\log p(x)$

We would like to maximize this in terms W. By taking derivatives and using the fact (from the first set of notes) that $\nabla_W |W| = |W|(W^{-1})^T$, we easily derive a stochastic gradient ascent learning rule. For a training example $x^{(i)}$, the update rule is:

$$W := W + \alpha \left(\begin{bmatrix} 1 - 2g(w_1^T x^{(i)}) \\ 1 - 2g(w_2^T x^{(i)}) \\ \vdots \\ 1 - 2g(w_d^T x^{(i)}) \end{bmatrix} x^{(i)T} + (W^T)^{-1} \right),$$

ICA Algorithm

$$W := W + \alpha \left(\begin{bmatrix} 1 - 2g(w_1^T x^{(i)}) \\ 1 - 2g(w_2^T x^{(i)}) \\ \vdots \\ 1 - 2g(w_d^T x^{(i)}) \end{bmatrix} x^{(i)T} + (W^T)^{-1} \right),$$

where α is the learning rate.

After the algorithm converges, we then compute $s^{(i)} = Wx^{(i)}$ to recover the original sources.

ICA Algorithm

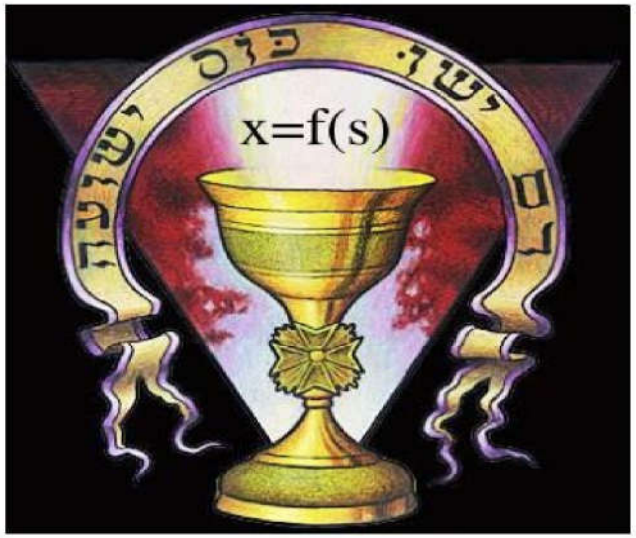
$x^{(n)} \ldots x^{(1)}$ تعداد نمونه‌ها که تعداد نمونه‌ها زیاد می‌شود

$n \uparrow$

Remark. When writing down the likelihood of the data, we implicitly assumed that the $x^{(i)}$'s were independent of each other (for different values of i; note this issue is different from whether the different coordinates of $x^{(i)}$ are independent), so that the likelihood of the training set was given by $\prod_i p(x^{(i)}; W)$. This assumption is clearly incorrect for speech data and other time series where the $x^{(i)}$'s are dependent, but it can be shown that having correlated training examples will not hurt the performance of the algorithm if we have sufficient data. However, for problems where successive training examples are correlated, when implementing stochastic gradient ascent, it sometimes helps accelerate convergence if we visit training examples in a randomly permuted order. (I.e., run stochastic gradient ascent on a randomly shuffled copy of the training set.)

ICA Generalizations

- Independent Subspace Analysis
- Multilinear ICA
- Blind Source Deconvolution
- Blind SubSpace Deconvolution
- Nonnegative ICA
- Sparse Component Analysis
- Slow Component Analysis
- Noisy ICA
- Undercomplete, Overcomplete ICA
- Varying mixing matrix
- Online ICA
- (Post) Nonlinear ICA



The Holy Grail

Algorithms

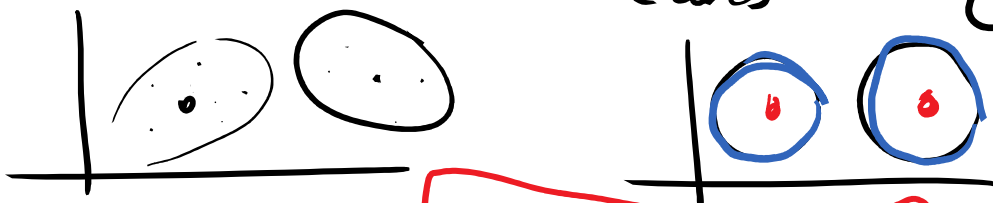
There are more than 100 different ICA algorithms…

- Mutual information (MI) estimation
  - Kernel-ICA [Bach & Jordan, 2002]
- Entropy, negentropy estimation
  - Infomax ICA [Bell & Sejnowski 1995]
  - RADICAL [Learned-Miller & Fisher, 2003]
  - FastICA [Hyvarinen, 1999]
  - [Girolami & Fyfe 1997]
- ML estimation
  - KDICA [Chen, 2006]
  - EM-ICA [Welling]
  - [MacKay 1996; Pearlmutter & Parra 1996; Cardoso 1997]
- Higherher order moments, cumulants based methods
  - JADE [Cardoso, 1993]
- Nonlinear correlation based methods
  - [Jutten and Herault, 1991]

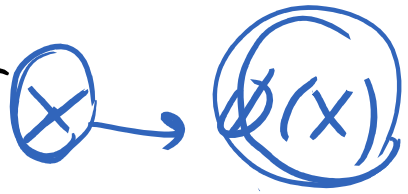
جمع بندی الگوریتم‌های unsupervised learning :

۱- روش‌های خوشه‌بندی

GMM - Kmeans



Kernel kmeans



Hierarchical clustering

DBSCAN (شکل دلخواه خوشه‌ها)

۲- روش‌های کاهش بعد

Autoencoder

کاهش بعد یا حذف نویز مناسب برای داده‌ها denoising

LDA → KDA

PCA → KPCA

non_linear PCA

Manifold learning:

مجموعه‌ای از روش‌های یادگیری non-parametric

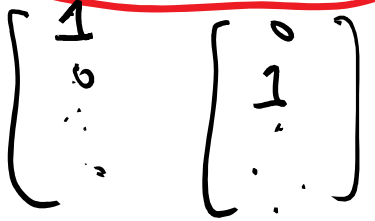
برای هر نمونه به صورت جزا نمونه‌ها یاد گرفته بدست می‌آید و می‌توان

یک مدل عمومی برای یادگیری چگونگی نمونه‌ها داشت.

Isomap

LLE





* با استفاده از یک embedding مناسب برای داده‌ها

۳- روش‌های جداسازی منابع:

$X = AS$

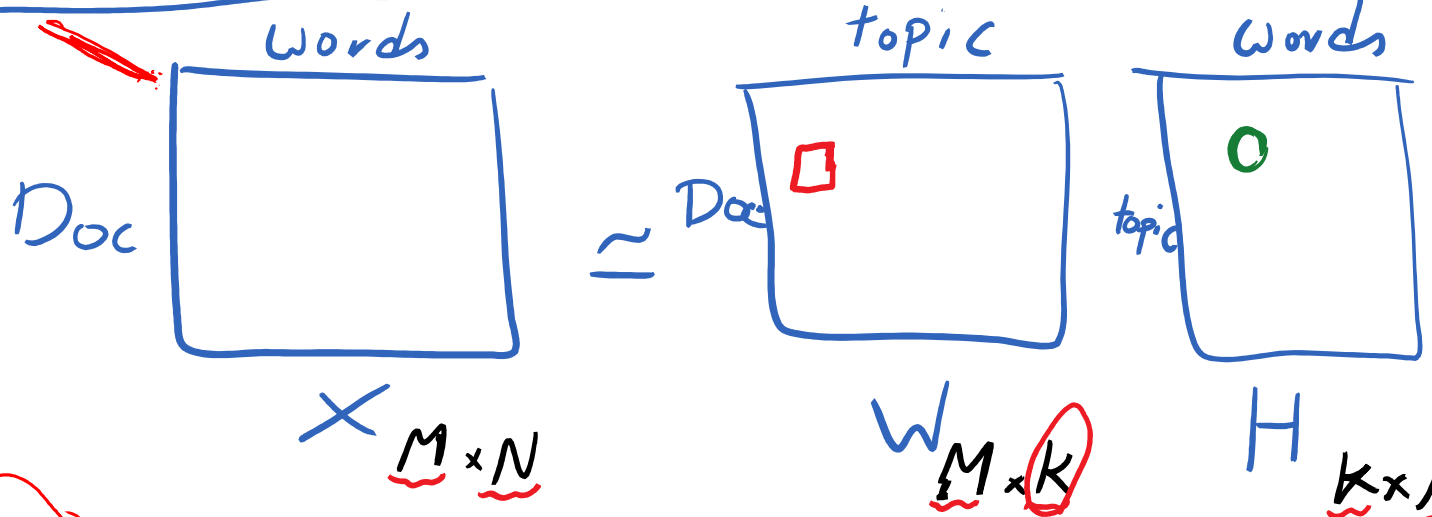
ICA

NMF

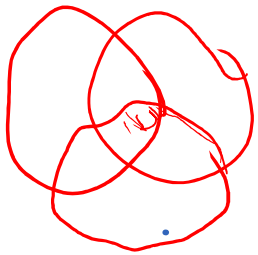
$X \simeq WH$

معیار فاصله - IS - KL - (اقلیدسی)

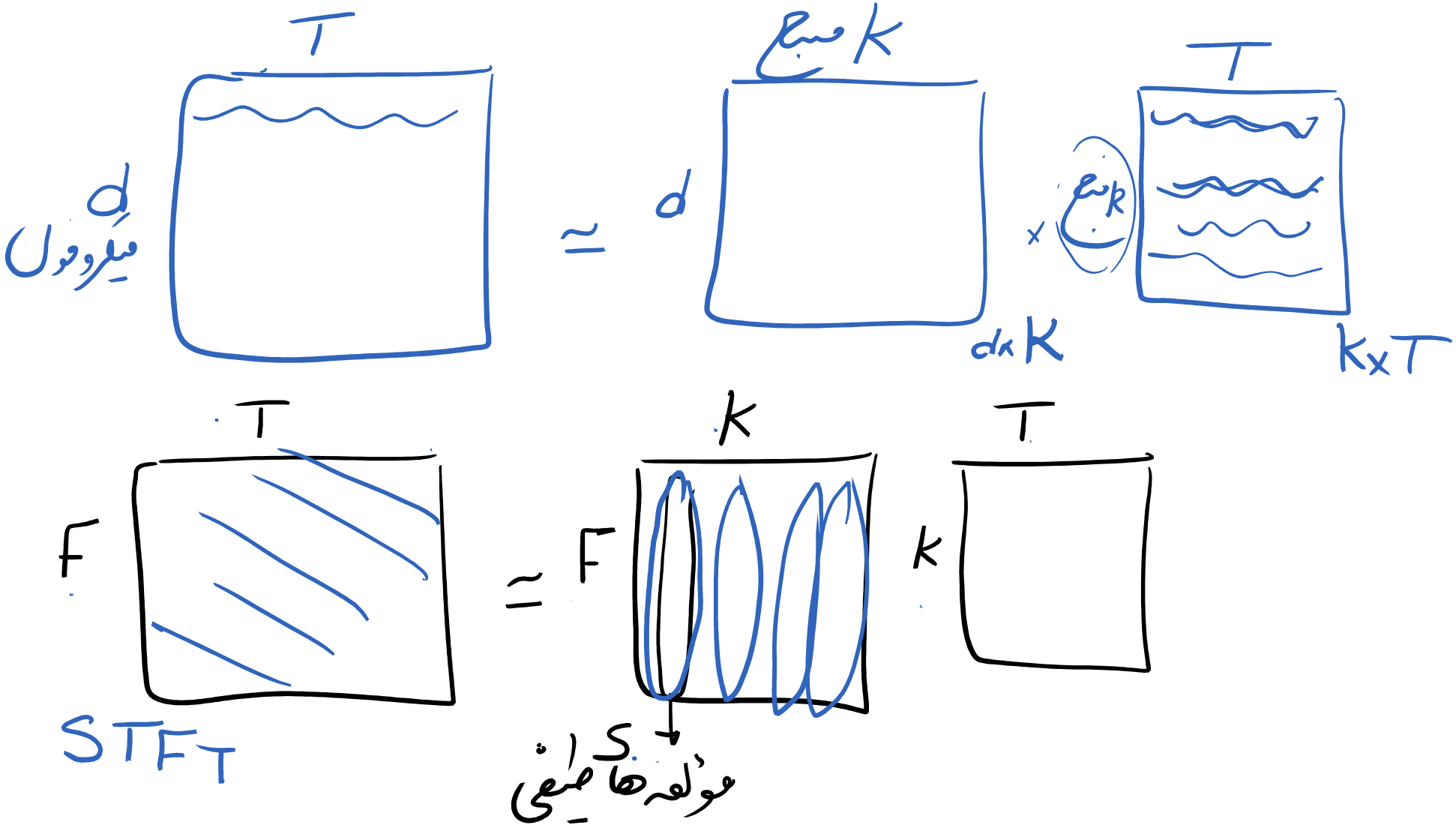
Bayesian NMF



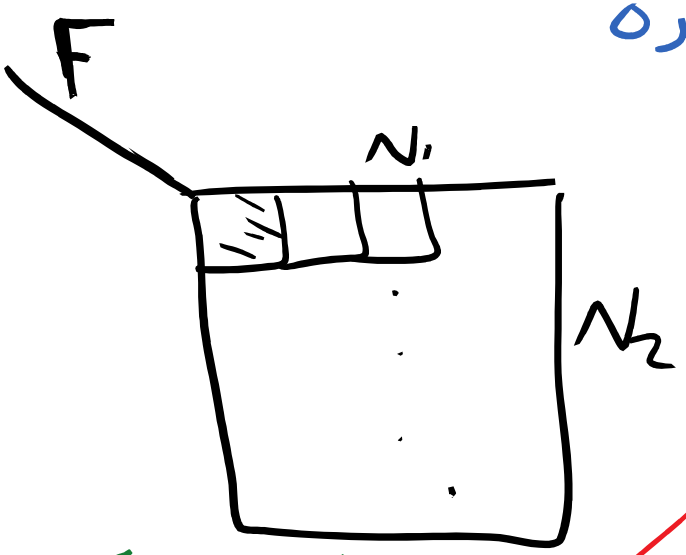
$X_{M \times N}$ $W_{M \times K}$ $H_{K \times N}$



NMF برای خوشه‌بندی هم می‌تواند به کار رود. Overlapped clusters


T
d
میلرودوں
≃
d
صبح k
dx K
x
صبح k
T
k x T
T
F
STF T
≃
F
k
T
k
S.
مولفه‌ها ضعفی

تجزیه تصاویر Hyperspectral به اجزای سازنده



$$F \times N_1 \times N_2$$

روش‌های تجزیه ماتریس
یا تنسور با استفاده
از نشانه‌های تصویر

